Actual growing norm input budgets

Prime norm study, 14 November 2026

Source freeze `0a12a195c66785b5b8b4b4409ab0cc02ee991f64`

Result. The literal coefficient polynomial of the computed full membership remainder obeys

$$\operatorname{wdeg}(R_j) \le 2\deg(P) - 2j,$$

with natural subtraction at zero. The theorem uses the actual multivariate coefficient support, for every arity $a = h + 2$, $h \in \mathbb{N}$. PCC-20261114-253550/2c4a15; unsigned [pcc:unverified]. This is an input bound. It is not a bound on full iterated ideal closure.

The actual family. Over a field K, retain the variables A_j $(1 \le j < a)$ and B_j $(0 \le j \le a-2)$, and public scalars α, β, r, z. Set

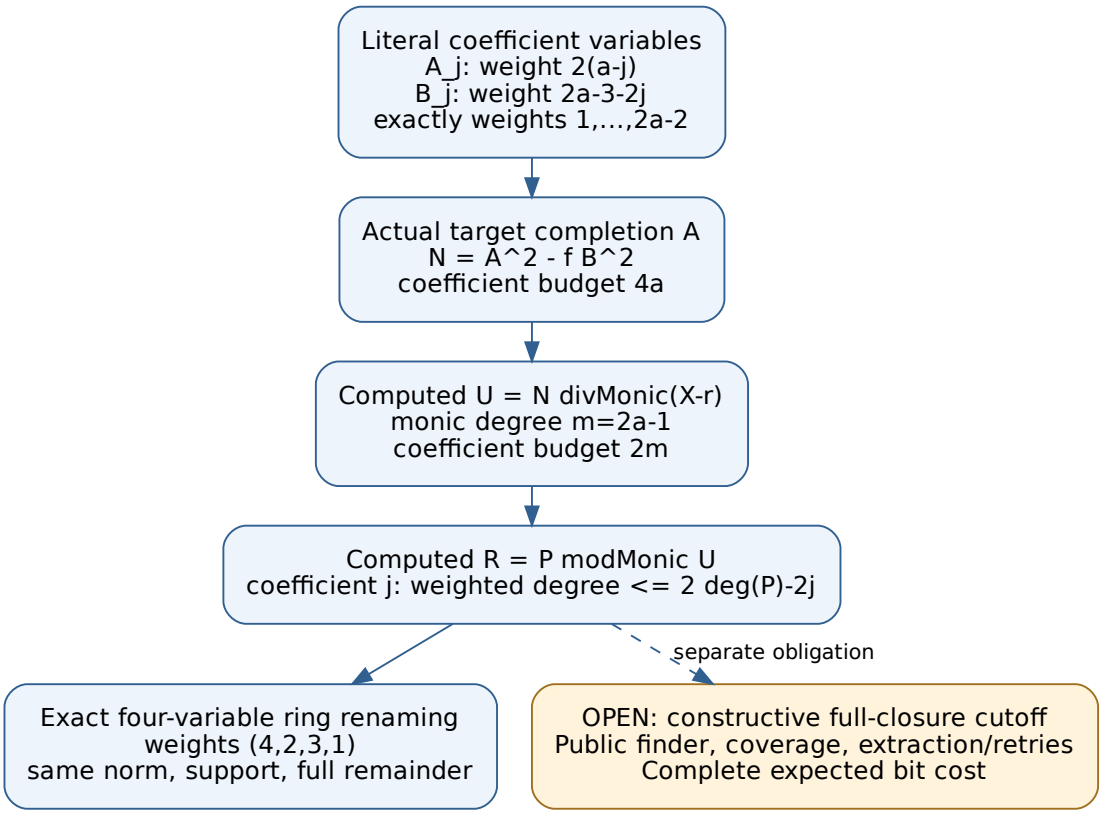
$$\begin{aligned}
D(X) &= X^a + \sum_{j=1}^{a-1} A_j X^j, & B(X) &= \sum_{j=0}^{a-2} B_j X^j,\\
A(X) &= D(X) + zB(r) - D(r), & f(X) &= X^3 + \alpha X + \beta,\\
N(X) &= A(X)^2 - f(X)B(X)^2, & U(X) &= N(X) \operatorname{divMonic} (X - r),\\
R(X) &= P(X) \operatorname{modMonic} U(X).
\end{aligned}$$

The public polynomial $P \in K[X]$ is arbitrary. A point-list nodal product is a possible application; its generation is not assumed free. The assigned weights are $w(A_j) = 2(a - j)$ and $w(B_j) = 2a - 3 - 2j$. An explicit bijection identifies them with $1, \dots, 2a - 2$; PCC-20261114-cad062 [pcc:unverified].

Support and incidence. The actual norm is monic of degree $2a$; its computed support U is monic of degree $m = 2a - 1$. These statements do not require a successful relation. If $z^2 = r^3 + \alpha r + \beta$, then $(X - r)U = N$ by exact target incidence; PCC-20261114-6c253b [pcc:unverified].

Division proof. A budget d means $w(e) + 2j \le d$ for every exponent e in the actual support of coefficient j. The norm has budget $4a$, and U has budget $2m$. The generic theorem follows Mathlib's actual well-founded monic-division recursion: each leading-term subtraction preserves the budget; the computed quotient has budget $d - 2m$, and the computed remainder has budget d. A public degree-B polynomial has budget $2B$, giving the displayed result. No quotient, normal form, moment sequence or successful support is supplied. PCC-20261114-3a3e7b/253550 [pcc:unverified].

Exact original-chart binding. At $h = 1$, an equivalence renames the coefficient polynomial ring to four indeterminates. The actual A, B, N, U and full remainder equal the existing four-variable source expressions, with weights $(4, 2, 3, 1)$. This is a polynomial-ring identity, not equality only on finite-field points. PCC-20261114-3a3330/69b0f2 [pcc:unverified].



Validation and provenance. Forty authored theorems in:
`lean/NormInputBudget.lean`
`lean/NormGrowingInput.lean`.

Final source build006 exits0 without warnings; failed drafts and raw logs are retained. Gate/fresh receipts and unsigned claim hashes are recorded in the companion Markdown report. No signed verification is asserted.

Sources checked(KB): unavailable. Primary fallback read the actual Mathlib support/weight lemmas and well-founded division recursion, plus existing NormMembershipElimination, NormPoleTransport and NormTargetFiber. Earlier weighted-GB and P1514/P1536 audits distinguish input grading from regularity, available normal forms and efficient public construction. No novelty claim is made for generic weighted division.

Next dependency. A constructive small full-closure cutoff for this growing family is still required. The prior finite weighted count can price a given cutoff; this input bound does not supply that cutoff. Public base coverage, signed output, field extraction, failed attempts and complete expected bit cost remain charged obligations. No normalized-row degree29 or original $W_{\mathrm{new}} = W_{\mathrm{base}} + CH$ jet map is established here. No solver, native fixture, paid job, timing measurement or Python cryptomath was run. Canonical performance graphs stay unchanged.